

Euclidean geometry – 1

Vectors, angles, simplices and volumes in $\mathbb{R}^n$

Homework

24 сентября 2026 года

Theoretical minimum. The Gram matrix $\Gamma(v_1, \dots, v_k) = (\langle v_i, v_j \rangle)$ is positive semidefinite, of the rank of the system, and $\sqrt{\det \Gamma}$ is the k -volume of the parallelepiped on $v_1, \dots, v_k$; the volume of an n -simplex $A_0 \dots A_n$ is $\frac{1}{n!} \sqrt{\det \Gamma(A_1 - A_0, \dots, A_n - A_0)}$ and, through $2\langle A_i - A_0, A_j - A_0 \rangle = d_{0i}^2 + d_{0j}^2 - d_{ij}^2$, a function of the distances (Cayley–Menger). By Cauchy–Binet the squared k -volume of a piece of a k -plane is the sum of the squared volumes of its projections onto the coordinate k -planes. Barycentric coordinates of X are the ratios $\pm v(S_i)/v(S)$; the centroid is $G = \frac{1}{n+1} \sum A_i$, and $\sum |XA_i|^2 = (n+1)|XG|^2 + \sum |GA_i|^2$. At most $2n$ non-zero vectors of $\mathbb{R}^n$ have pairwise non-acute angles and at most $n+1$ pairwise obtuse ones; at most $\frac{n(n+1)}{2}$ lines through the origin are equiangular, because the projectors $v_i v_i^T$ are linearly independent. Points are in general position if no $n+1$ of them lie in a hyperplane.

1. Let $A_0 A_1 \dots A_n$ be a regular n -simplex with edge a . Prove that its circumradius and inradius are $R = a \sqrt{\frac{n}{2n+2}}$ and $r = R/n$, that its volume is $\frac{a^n}{n!} \sqrt{\frac{n+1}{2^n}}$, that the vectors from the centre to two vertices make the angle $\arccos(-\frac{1}{n})$, and that the dihedral angle between two facets equals $\arccos \frac{1}{n}$.
2. Prove the Leibniz formula $\sum_{i=0}^n |XA_i|^2 = (n+1)|XG|^2 + \sum_{i=0}^n |GA_i|^2$ and deduce that for an n -simplex with circumcentre O , circumradius R and centroid G

$$|OG|^2 = R^2 - \frac{1}{(n+1)^2} \sum_{0 \leq i < j \leq n} |A_i A_j|^2.$$

Conclude that $\sum_{i < j} |A_i A_j|^2 \leq (n+1)^2 R^2$, with equality exactly when

$O = G$; for instance, the squares of the six edges of a tetrahedron inscribed in a unit sphere sum to at most 16.

3. Show that the six diagonals of a regular icosahedron are equiangular lines in $\mathbb{R}^3$ with the angle $\arccos \frac{1}{\sqrt{5}}$, and that the 28 vectors of $\mathbb{R}^8$ having two coordinates equal to 3 and six equal to -1 span 28 equiangular lines in a 7-dimensional subspace. Thus the bound $\frac{n(n+1)}{2}$ is attained for $n = 2, 3, 7$.
4. Let $A_1, \dots, A_n$ be points of an ellipsoid with centre O in $\mathbb{R}^n$ such that OA_i , for $i = 1, \dots, n$, are mutually orthogonal. Prove that the distance of the point O from the hyperplane $A_1 A_2 \dots A_n$ does not depend on the choice of the points $A_1, \dots, A_n$. (VJIMC 1999)
5. Determine all positive integers N for which the sphere $x^2 + y^2 + z^2 = N$ has an inscribed regular tetrahedron whose vertices have integer coordinates. (Putnam 2021 A3)
6. For which positive integers n is there an $n \times n$ matrix with integer entries such that every dot product of a row with itself is even, while every dot product of two different rows is odd? (Putnam 2011 A4)
7. Show that there do not exist four points in the Euclidean plane such that the pairwise distances between the points are all odd integers. (Putnam 1993 B5)
8. Let $v_1, \dots, v_m \in \mathbb{R}^n$ be unit vectors. Prove that the matrix $(n(v_i \cdot v_j)^2 - 1)_{i,j=1}^m$ is positive semidefinite. Deduce that if m lines through the origin of $\mathbb{R}^n$ make pairwise angles $\arccos \alpha$ with $n\alpha^2 < 1$, then $m \leq \frac{n(1-\alpha^2)}{1-n\alpha^2}$, and check that this bound is attained by the examples of problem 3. (МФТИ 2021)
9. Let B be a set of more than $2^{n+1}/n$ distinct points with coordinates of the form $(\pm 1, \pm 1, \dots, \pm 1)$ in n -dimensional space with $n \geq 3$. Show that there are three distinct points in B which are the vertices of an equilateral triangle. (Putnam 2000 B6)
10. We say that a subset of $\mathbb{R}^n$ is k -almost contained by a hyperplane if there

are less than k points in that set which do not belong to the hyperplane. We call a finite set of points *k-generic* if there is no hyperplane that k -almost contains the set. For each pair of positive integers k and n , find the minimal number $d(k, n)$ such that every finite k -generic set in $\mathbb{R}^n$ contains a k -generic subset with at most $d(k, n)$ elements.

(IMC 2014)